

NeuroStrike: Neuron-level Attacks on Aligned LLMs

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Introduction to LLMs

- Large Language Models (LLMs) are advanced Artificial Intelligence systems trained using huge datasets collected from books, websites, articles, and code repositories.
- They are based on transformer neural network architectures that process and generate human-like text.
- LLMs can perform multiple tasks such as text generation, translation, coding assistance, summarization, and question answering.
- Popular examples include GPT, Gemini, Claude, and Llama.
- These models are increasingly used in industries like healthcare, education, finance, and cybersecurity.

What is AI Alignment?

- AI alignment ensures that AI systems behave according to human values and safety guidelines.
- Alignment methods reduce harmful, biased, or dangerous outputs.
- Reinforcement Learning from Human Feedback (RLHF) is commonly used.
- Safety fine-tuning teaches models to reject unsafe prompts.
- Aligned models are expected to provide ethical and secure responses.

Problem Statement

- Even aligned LLMs may contain hidden vulnerabilities.
- Traditional attacks mainly focus on prompt engineering or jailbreak techniques.
- Researchers questioned whether internal neural components can be exploited.
- This led to the development of NeuroStrike.
- The research investigates whether alignment can be bypassed at the neuron level.

What is NeuroStrike?

- NeuroStrike is a neuron-level attack framework for aligned LLMs.
- It targets internal neurons responsible for safety and refusal behavior.
- Instead of changing prompts, it manipulates hidden neural activations.
- The attack demonstrates weaknesses in current AI safety mechanisms.
- It highlights that safety may depend on a very small number of neurons.

Neurons in Transformer Models

- Transformer models contain millions or billions of neurons.
- Different neurons learn different semantic and behavioral patterns.
- Some neurons contribute to factual reasoning.
- Others contribute to toxicity filtering and refusal generation.
- NeuroStrike specifically identifies neurons linked to alignment behavior.

Core Idea Behind the Attack

- The researchers monitor neuron activation during safe and unsafe responses.
- Safety-related neurons show strong activation during refusals.
- Attackers suppress or manipulate these neurons.
- Once modified, the model may stop refusing harmful requests.
- This creates unsafe outputs even in aligned systems.

Attack Workflow

- Step 1: Identify alignment-sensitive neurons.
- Step 2: Record activation patterns during refusal responses.
- Step 3: Apply targeted neuron intervention.
- Step 4: Run inference with manipulated neurons.
- Step 5: Measure changes in harmful response generation.

Neuron Identification Techniques

- Activation analysis is used to track neuron behavior.
- Researchers compare activations for harmful and harmless prompts.
- Importance scoring helps identify critical neurons.
- Gradient-based analysis reveals influential components.
- Only a small subset of neurons may control safety behavior.

Key Findings

- A very small number of neurons significantly affect alignment.
- Suppressing these neurons weakens refusal behavior.
- Models generated harmful outputs after manipulation.
- Alignment robustness was lower than expected.
- The study proved that neuron-level vulnerabilities exist.

Examples of Attack Impact

- Aligned models may answer restricted or unsafe questions.
- Toxicity filters become less effective.
- The model may reveal harmful instructions.
- Content moderation systems can fail after manipulation.
- Safety guarantees become unreliable.

Security Implications

- Internal access to model activations becomes a major security concern.
- AI systems deployed in sensitive environments may be vulnerable.
- Attackers could bypass safeguards without changing prompts.
- This threatens trust in aligned AI systems.
- The research emphasizes the importance of robust AI security.

Advantages of NeuroStrike Research

- Provides deeper understanding of internal LLM behavior.
- Encourages development of more robust alignment methods.
- Improves AI interpretability research.
- Helps researchers identify weak points in neural architectures.
- Supports the creation of safer future AI systems.

Limitations of the Study

- The attack requires internal model access.
- Results may vary across different architectures.
- Interpretability of neural networks remains difficult.
- Defenses against neuron-level attacks are still limited.
- Scaling analysis for very large models is computationally expensive.

Results

Future Research Directions

- Designing robust neuron-level defense systems.
- Understanding hidden representations inside transformers.
- Improving transparency of AI decision making.
- Building self-monitoring AI architectures.
- Combining interpretability with cybersecurity research.

Conclusion

- NeuroStrike demonstrates that alignment can be bypassed internally.
- Neuron-level attacks reveal critical weaknesses in current LLM safety methods.
- The research highlights the need for robust, interpretable, and secure AI systems.
- Future AI safety must address vulnerabilities inside neural architectures.
- Understanding neuron behavior is essential for trustworthy artificial intelligence.

References

- **NeuroStrike: Neuron-Level Attacks on Aligned LLMs(USENIX 2026)**
- <https://github.com/wu-lichao/NeuroStrike-Neuron-Level-Attacks-on-Aligned-LLMs>